Learning Hyperplane Branching for Integer Optimization

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Integer Optimization

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✓ Dilkina et al., (2016,2017) : learning to branch

HYPERPLANE BRANCHING

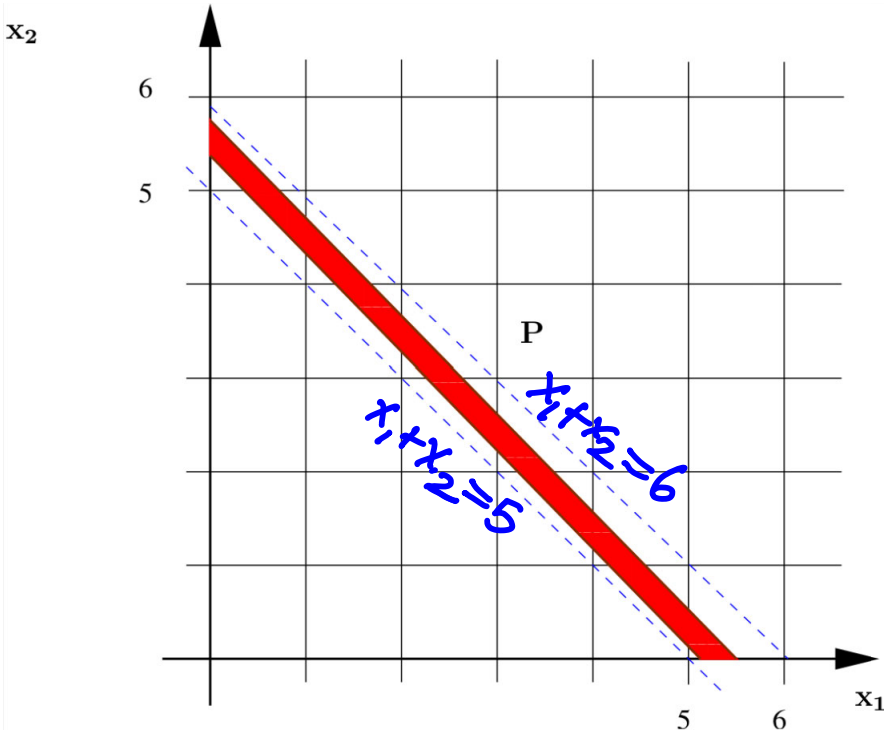
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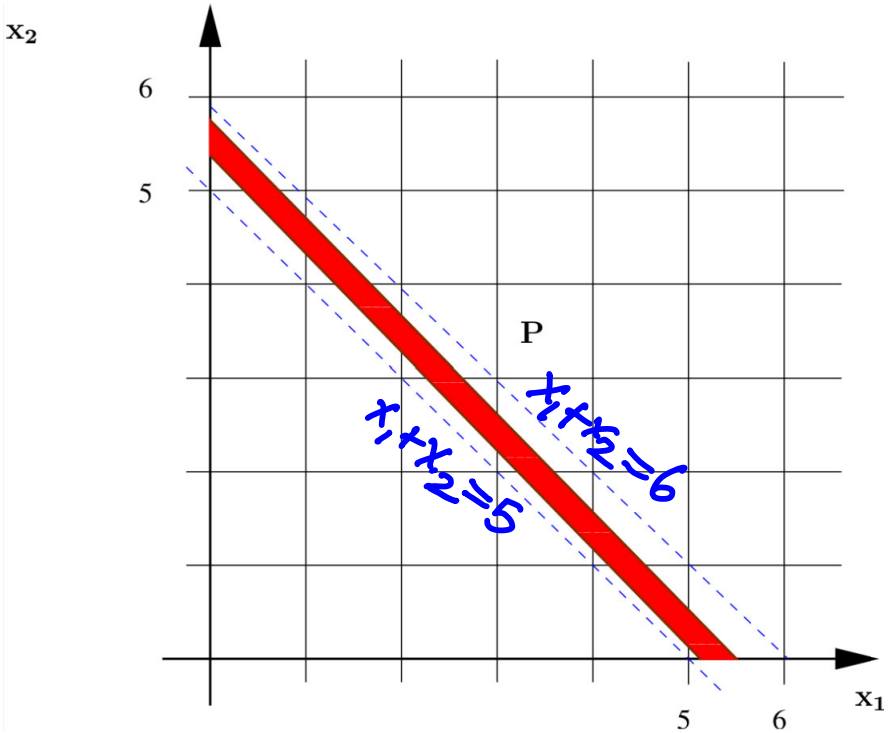


$$(P)\left\{\beta' \le \bar{a}^T\bar{x} \le \beta \atop 0 \le \bar{x} \le \bar{u}\right\},\quad P \cap \mathbb{Z}^n = \emptyset$$

$$\bar{a} = \bar{p}M + \bar{r} \quad (M: \text{large})$$

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✓ branching on $\bar{p}^T\bar{x}$ kills the problem!

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? DKP learner on general IPs ($A\bar{x} \leq \bar{b}$)?

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- ✓ CKP with $\bar{a} = \bar{p}_1 M_1 + \cdots + \bar{p}_k M_k + \bar{r}$
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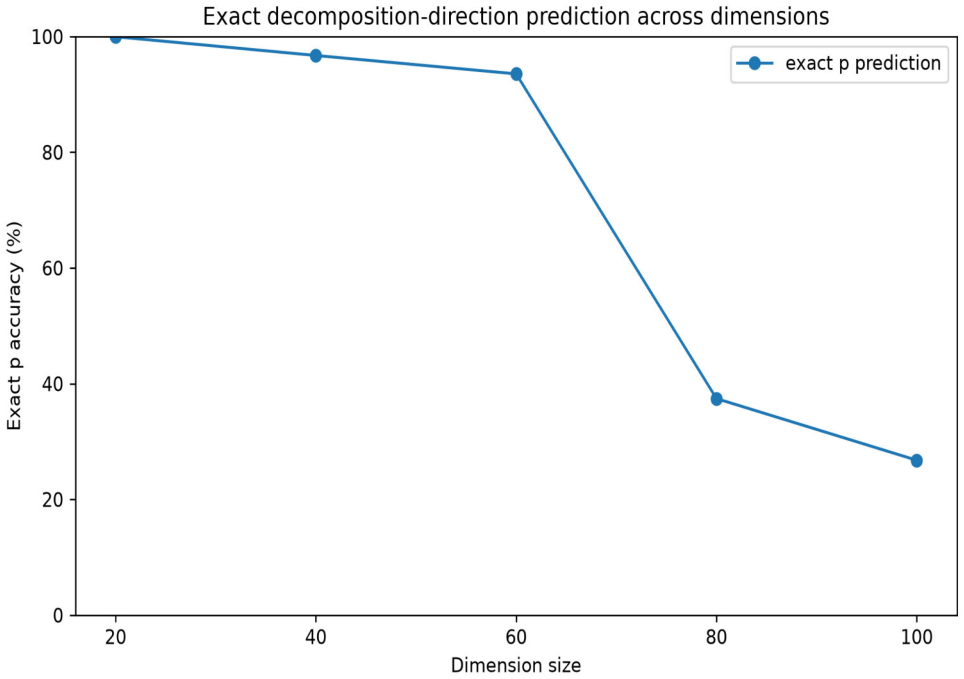
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  - study DKP learner recovering $\bar{F}_1, \ldots, \bar{F}_k$
- ✓ Generalize to $A\bar{x} \leq \bar{b}$
  - helps when DKP directions present
  - fails in general instances

DKP LEARNER

* NN for $(\bar{\alpha}, \bar{\beta}', \beta)$ $\xrightarrow{\text{predict}}$ $\bar{P}$ s.t. $\bar{\alpha} = \bar{P}M + \bar{r}$

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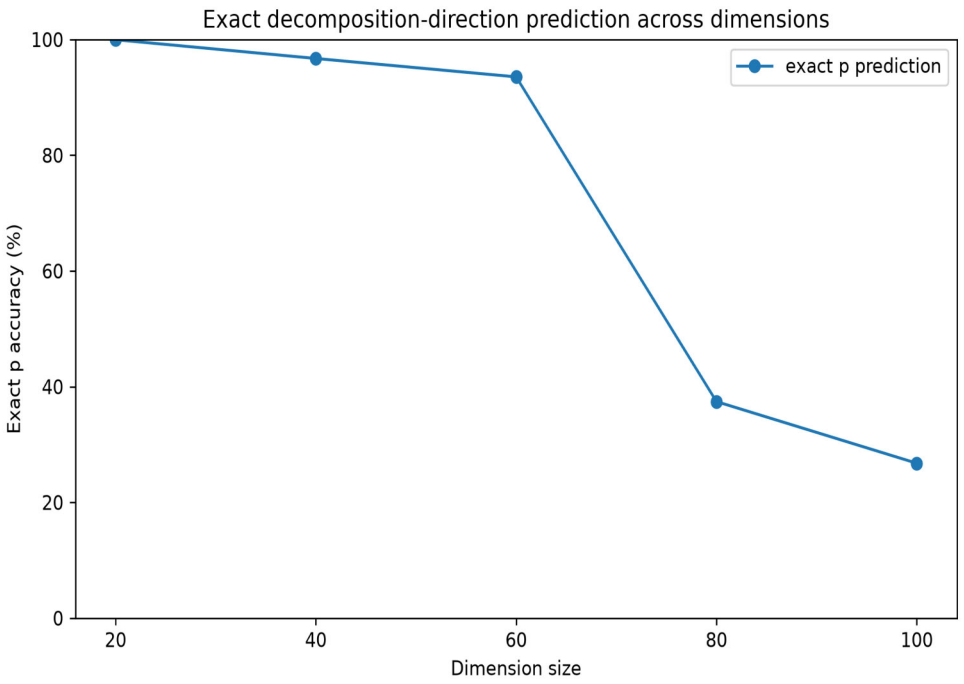


$n \rightarrow$

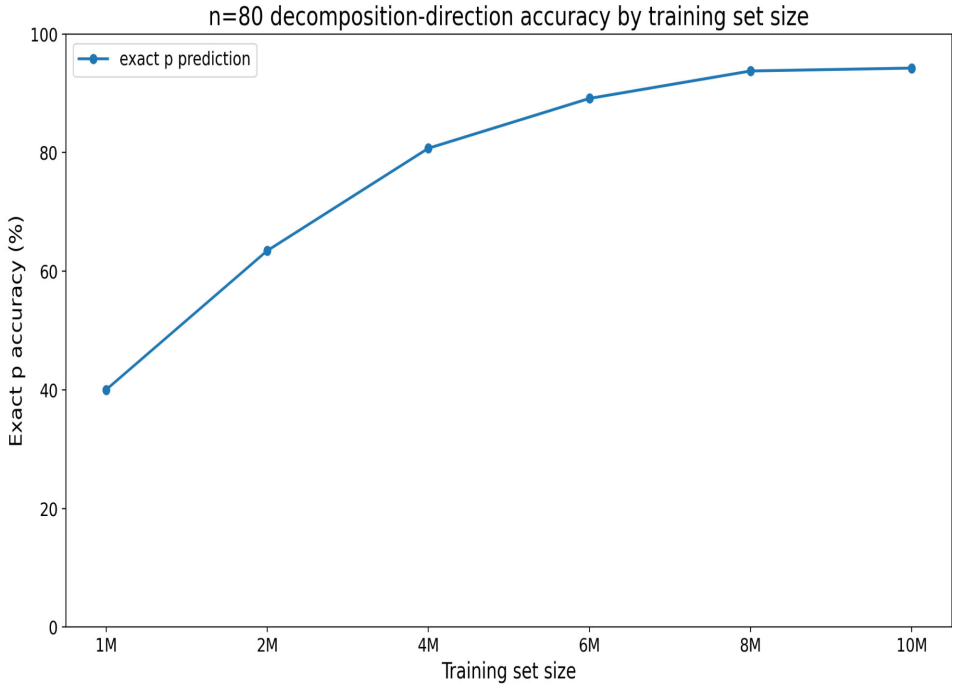
training set = 10^6 instances

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training set = 10^6 instances

$n = 80$

DKP CERTIFICATES

Lemma Let $S \subset \mathbb{R}^n$ and $\bar{p} \in \mathbb{Z}^n \setminus \{\bar{0}\}$ s.t. $\bar{p}^T \bar{x} \notin \mathbb{Z}$ for some $\bar{x} \in S$. Then $\exists M, \bar{r}, \beta', \beta$ s.t. DKP w/ $\bar{a} = \bar{p}M + \bar{r}$ has $S \cap \{\beta' \leq \bar{a}^T \bar{x} \leq \beta\} \neq \emptyset$.

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Theorem Let $\mathcal{F} \subset \mathbb{R}^n$ be bounded, $\mathcal{F} \neq \emptyset$ but $\mathcal{F} \cap \mathbb{Z}^n = \emptyset$. Let $P = [\bar{p}_1^T \cdots \bar{p}_r^T]$ s.t. $P(\mathcal{F}) \cap \mathbb{Z}^r = \emptyset$ and r is minimal. Then each $\bar{p}_i$ is DKP-representable

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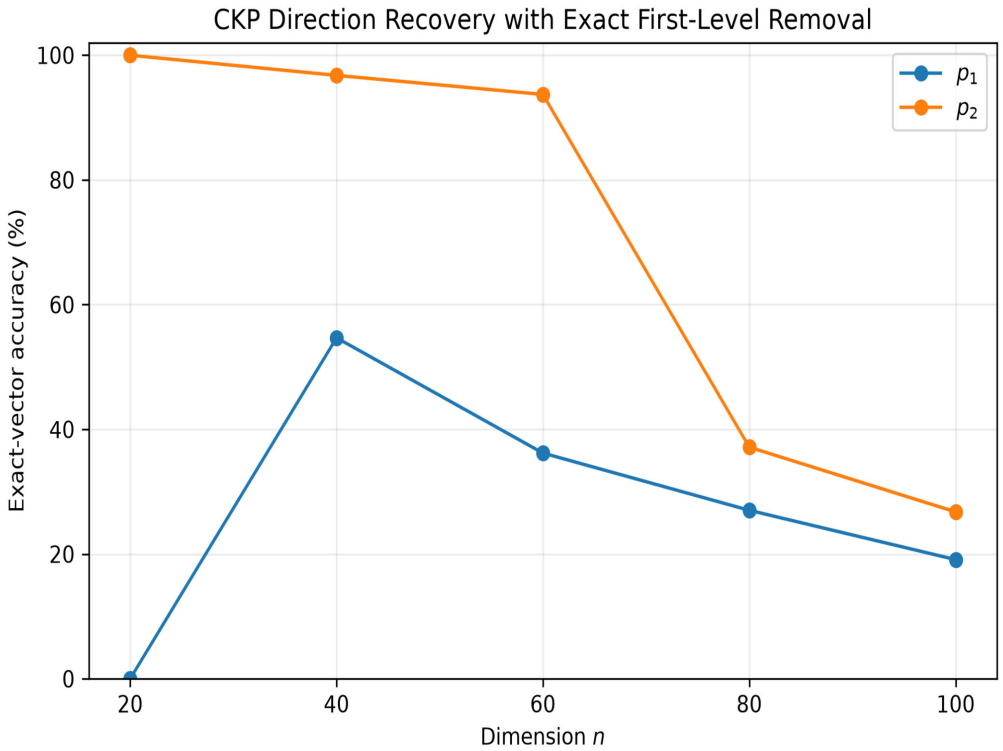
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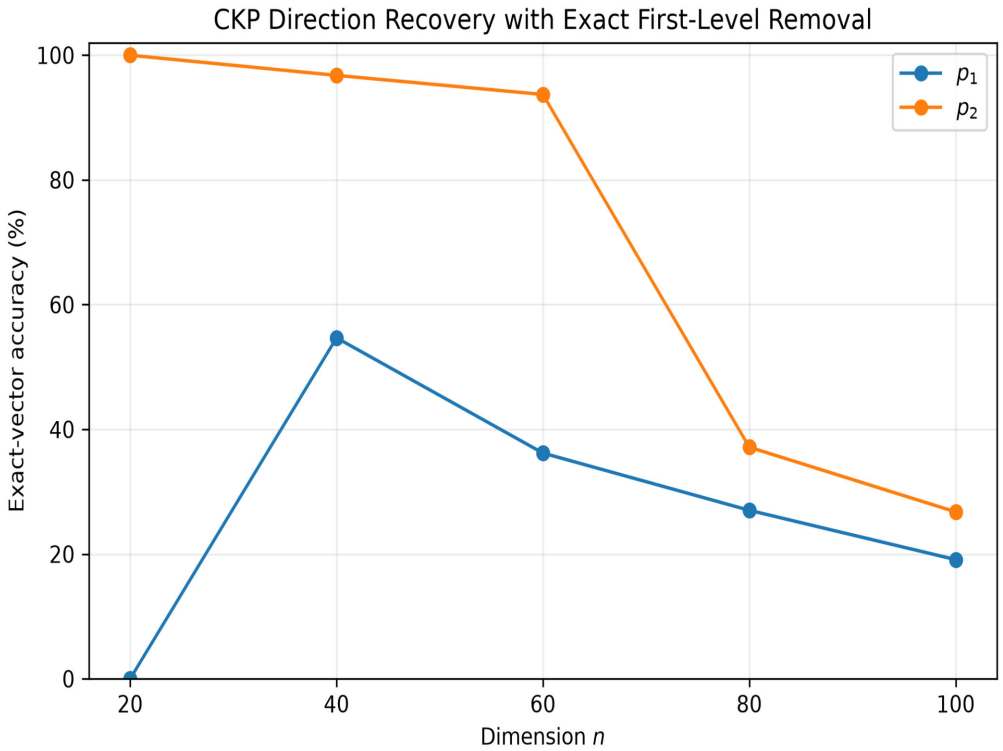
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? Can DKP learner predict $\bar{p}_1, \ldots, \bar{p}_{k-1}$ and $\bar{p}_k$?

CKPs with DKP Learner

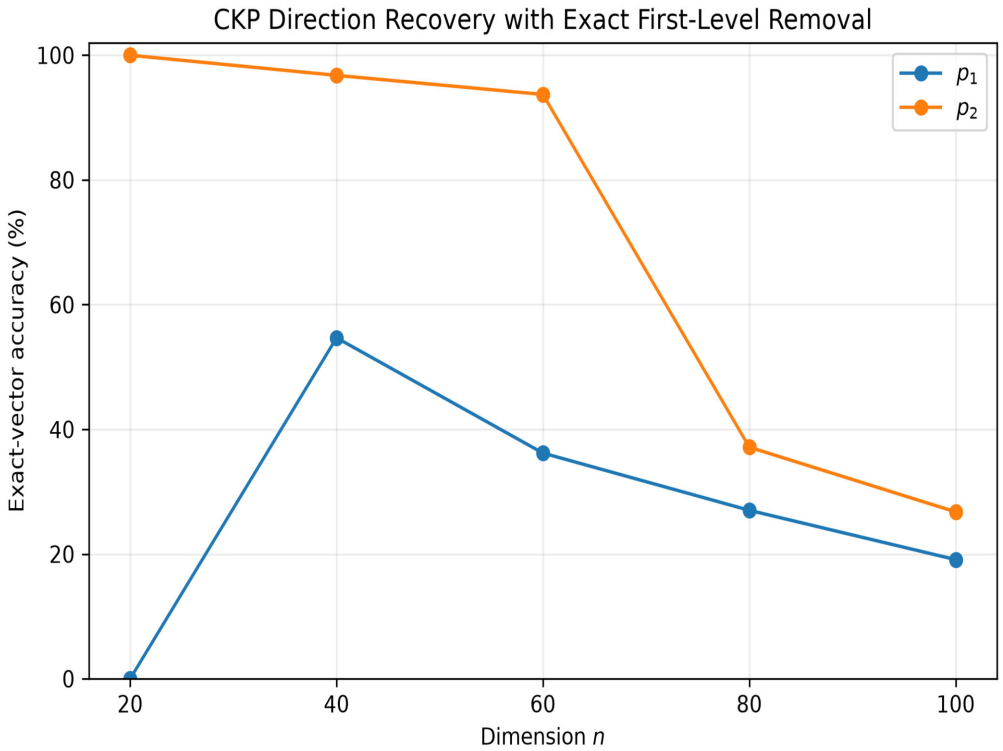


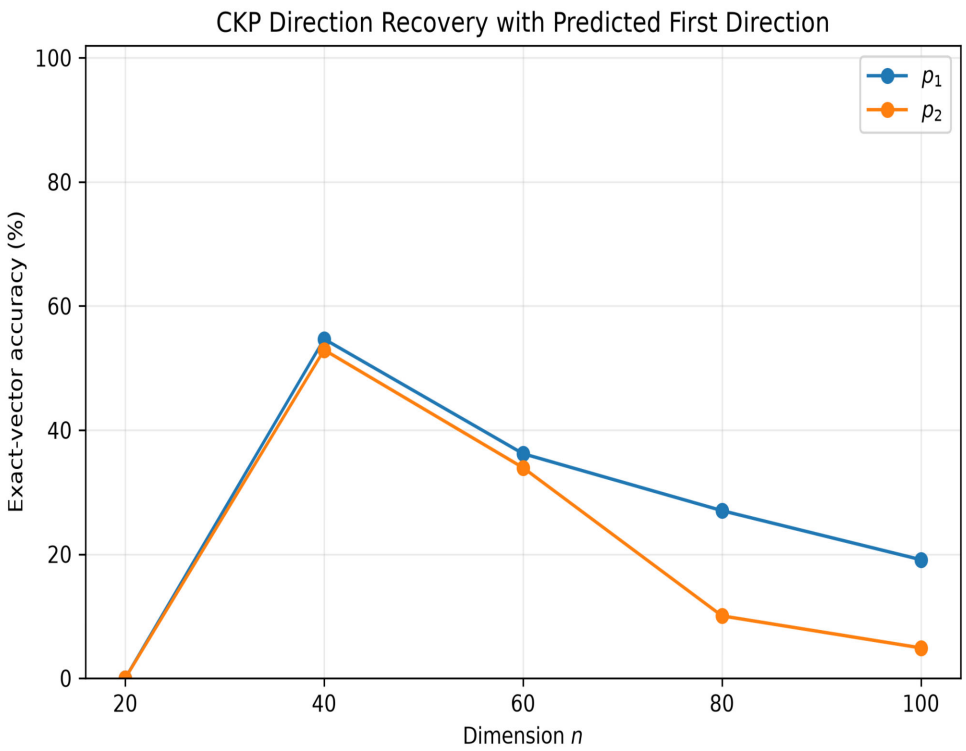
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CKPs WITH DKP LEARNER





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* struggles to predict $\bar{p}_2$ when predicted $\bar{p}_1$ is branched on

Marketshare Problems

* $\{\bar{x} \mid A\bar{x} = \bar{b}, 0 \leq \bar{x} \leq \bar{1}\} \cap \mathbb{Z}^n = \phi$, A: m rows

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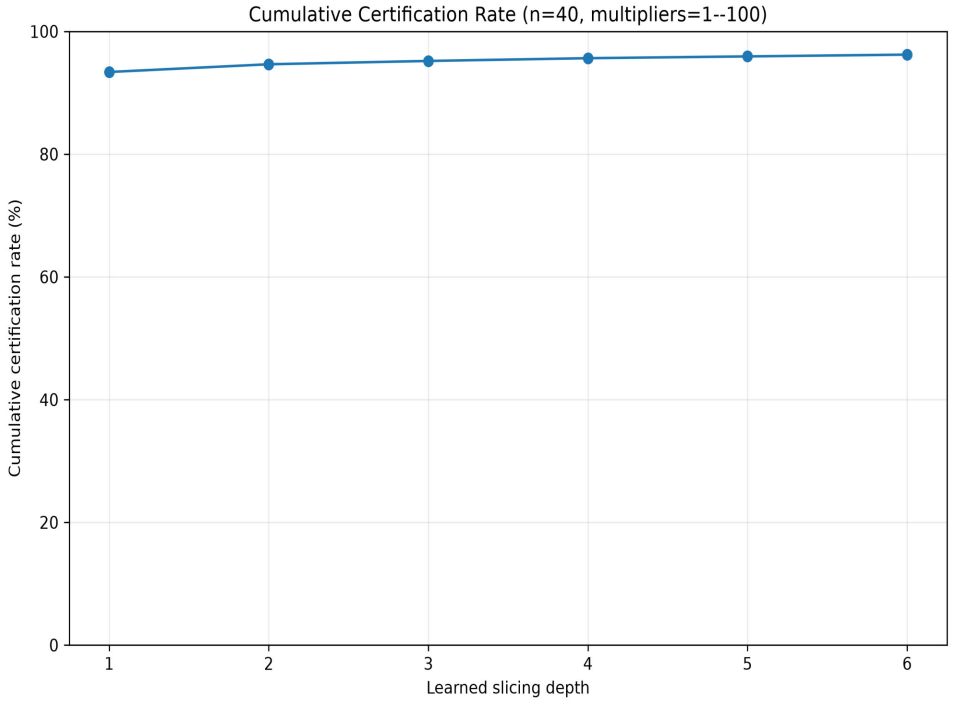
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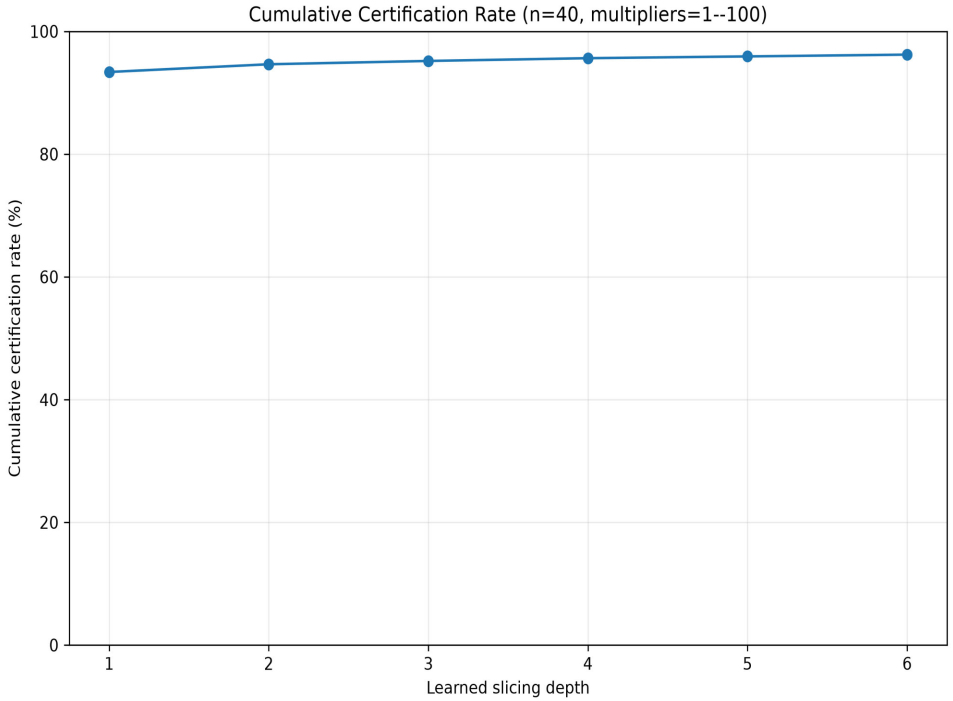
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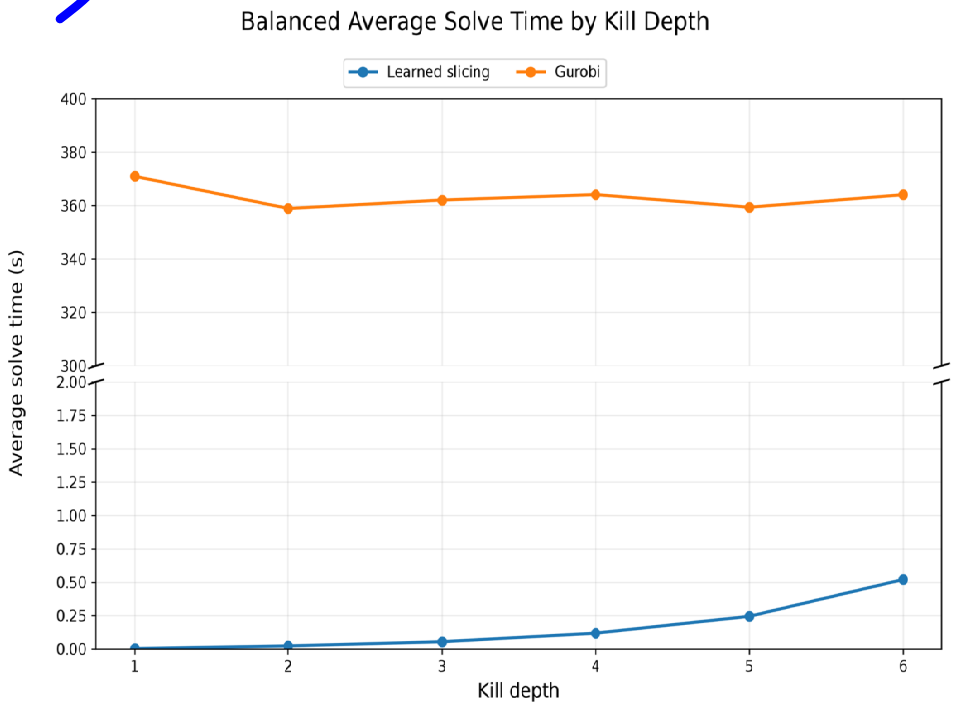
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v/s Gurobi

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